

How a Language Model Works

GLASS BOX · THREE 10-MINUTE ACTS · FOR EDUCATORS

WHAT THIS IS

A complete, working language model — 42,458 parameters of real transformer, the same architecture as the frontier models — that trains from scratch on the device in front of your audience. Act 1 shows what a language model *is*. Act 2 shows how **reasoning models** are made from one. Act 3 shows how a reasoning model plus tools becomes an **agent**. Nothing is faked: every number on screen is computed live, and each act ends with a "What is real here" box you can read aloud.

ACT 1 — THE PREDICTOR (10 MIN)

0:00

\$1.1 — type a sentence, watch it become numbered tokens.

"A model never sees words. It sees a list of numbers. Everything a chatbot has ever 'read' entered as a list like this one."

1:30

\$1.2 — tap a row of the parameter map.

The pixels are the actual weights, currently random noise. Show the scale bars below it.

"These 42,458 numbers are the entire machine. A frontier model is this exact picture with about fifty million times more pixels — and until it is trained, it is all static."

3:00

\$1.3 — read the untrained sample aloud.

Let it be ridiculous. Then press **Train** and narrate the loss falling while live samples stream.

"Its only teacher: guess character 41 from the last 40, be told the truth, nudge all 42,458 numbers a hair. Repeat."

Press train through each era — babble, words, phrases, sentences — reading one sample per era. The era comparison grid fills as you go. This is the demo's spine; give it time. When someone asks what it is reading, open "**Read its entire textbook**": an original 4,701-character story written for this demo, 252 distinct words — its whole universe. Everything it writes is a recombination of that page.

7:00

\$1.4 — the probability table.

Type a prompt, show the ranked guesses. Play with temperature at 0 (repetitive), 0.8, then 1.6 (unhinged). Switch back to the **newborn** era chip: same machine, useless numbers.

"This table is the only thing it ever does. Chat is this, ten thousand times in a row."

8:30

\$1.5 — tap characters in the attention view.

Gold = where it is looking. Try tapping the second "the" — heads often glance back at earlier occurrences and at the word after them.

"Attention is the invention that let each position rummage through everything before it. It is the T in GPT."

You need: one phone, tablet or laptop. Open `index.html`.
No install, no login, no network. Projecting? ⚙ **Settings** → **Presentation mode**.
Do not train before the session — the untrained model is the opening act.
Acts stand alone: teach one, two, or all three.

ACT 2 — THE REASONER (10 MIN)

0:00

Frame it before pressing anything:

two identical brains, equal training time; one sees only 347+285=632., the other sees the carries written out. Ask the room to predict the two exam scores. **Then press "Run the experiment"** and talk over the two minutes of training — there are live attempts and spot-checks to point at.

4:00

The reveal:

typically around half right versus nearly all right. Scroll to \$2.2 and read a red row from each side — Model D's misses are "close, right shape" (it learned the vibe of an answer); Model S's rare miss is one bad written step.

"Chain-of-thought isn't a trick prompt. It changes what is learnable: each small step is easy, the leap is not. 'Thinking' models are models trained to take the steps."

6:30

\$2.3 — press Run.

A deliberately under-trained sibling of Model S climbs as it gets more attempts, because a verifier checks the written steps — never the answer key. Show the accepted/rejected chains under the bars.

"Same frozen model on every row. We bought accuracy with compute at answer time. That is the 'thinking time' dial on every frontier reasoning model."

8:00

\$2.4 — press Self-improvement three times.

Watch accuracy climb round over round, and point out it now keeps far more of its own work than when it started (82 of 120 in round 1, over 100 by round 3).

"No new human data entered. It posed problems, kept what its verifier certified, and studied its own best work. Reinforcement learning — the essence of how reasoning models are actually trained."

ACT 3 — THE AGENT (10 MIN)

0:00

\$3.1 — read the loop out loud once

(context → model → tool → observation), then open the tool definitions fold.

"An agent is not a new kind of AI. It is the same predictor, in a loop, where some of its text makes things happen."

1:30

\$3.2 — "you are the model" first.

Hand the device to a volunteer and let the room argue over each choice. The tempting wrong choices really execute — picking the cheapest pizzeria without reading its hours is the lesson, not a mistake.

5:30

Press \$3.2's restart run, then "watch it run".

Two beats to point at: the calculator error (9 x \$13.00) and the recovery on the next turn — errors are just observations; and the context meter growing — every turn, the whole transcript is re-read.

8:00

The closer: set the window to "tiny", restart run, watch it run.

Mid-task, the goal falls out of the visible window and the agent stops dead, telling you why. Open "what the model sees" and show the red truncation line.

"The model remembers nothing between turns. The context window is its entire working memory — most of real agent engineering is a fight with exactly this."

MISCONCEPTIONS TO DRAW OUT

"It's understanding the story."

It is predicting the next character of one children's story. The interesting question — raise it, don't resolve it — is whether "understanding" at frontier scale is different in kind or only in degree. The demo gives both sides ammunition.

"It memorized the text."

42,458 parameters cannot hold 4,700 characters of text verbatim plus grammar; and its samples are recombinations, not quotes. (Frontier models *can* memorize; that is a real issue — different regime.)

"Reasoning models are just prompted to 'think step by step.'"

Prompting only elicits it. Act 2 shows the training version: the steps are in the data (§2.1), and reinforcement selects for chains that verify (§2.4). That is the difference between a chat model and a reasoning model.

"The agent decided to check the hours — so it wanted the right answer."

In this demo the decision rules are scripted (the page says so on screen). What is genuinely shown: the loop, real tools, real errors, and total dependence on context. Goals-talk is a description of behavior, not evidence of desire.

"Bigger context windows will fix agents."

They help, and real windows are ~200k+ tokens. But cost grows with context, attention gets diluted, and tasks can outlive any window — hence summarizing, note-taking (watch the agent's note tool), and retrieval.

"This is too small to say anything about ChatGPT/Claude."

Fair challenge — answer it honestly: what transfers is the mechanism (same architecture, same training rule, same loop), not the capability. Every "what is real here" box states exactly where the toy ends and scale begins.

TIMING & LOGISTICS

- Act 1 training totals ~2–3 minutes of compute across four presses; each burst streams live samples, so narrate over it. Act 2's experiment is one ~2–3 minute press; §2.3 ≈ 40s; each §2.4 round ≈ 20–30s.
- Training happens live but is **seeded and deterministic** — the engine computes its own math so the same run reproduces exactly, on every device (the shipped run: Model D ~54%, Model S ~96%, §2.3 climbs 60→78%, §2.4 rounds 49→77→83→84→85%). You can rehearse to the digit. The README documents the honest spread across other seeds.
- Phone works; a projected laptop is better for rooms. **Presentation mode** (⌕ Settings) enlarges everything one notch.
- **Controls:** ? opens the **Guide** overlay — it also opens on every load, so a late arrival can read it themselves. ⌕ **Settings** holds **Open Presenter Notes** (this document, on screen), **Presentation mode** and **Reset**.
- Refreshing the page forgets everything — which is itself a teachable moment, but not one you want at minute 25. **Settings** → **Reset** does the same on purpose, for a clean start between rooms; it leaves Presentation mode as you set it.

QUESTIONS YOU WILL GET

- **"How long would a frontier model take to train this way?"** This page does ~15 updates/s on ~43k parameters. Frontier training is ~10²³ floating-point operations — months on tens of thousands of GPUs. Same rule, different budget.
- **"Where do the words come from at temperature 0?"** Deterministic: always the top of the probability table. All variety in AI output is controlled dice-rolling over that table.
- **"Is the verifier cheating?"** It never sees the true answer — only whether the written steps are internally valid arithmetic that assembles into the stated answer. That is why it can reject a right answer with wrong work.
- **"What would make Act 3's agent 'real'?"** Replace the scripted rulebook with Act 2's model at scale — that is literally the architecture of Claude, ChatGPT agents, and coding assistants.